

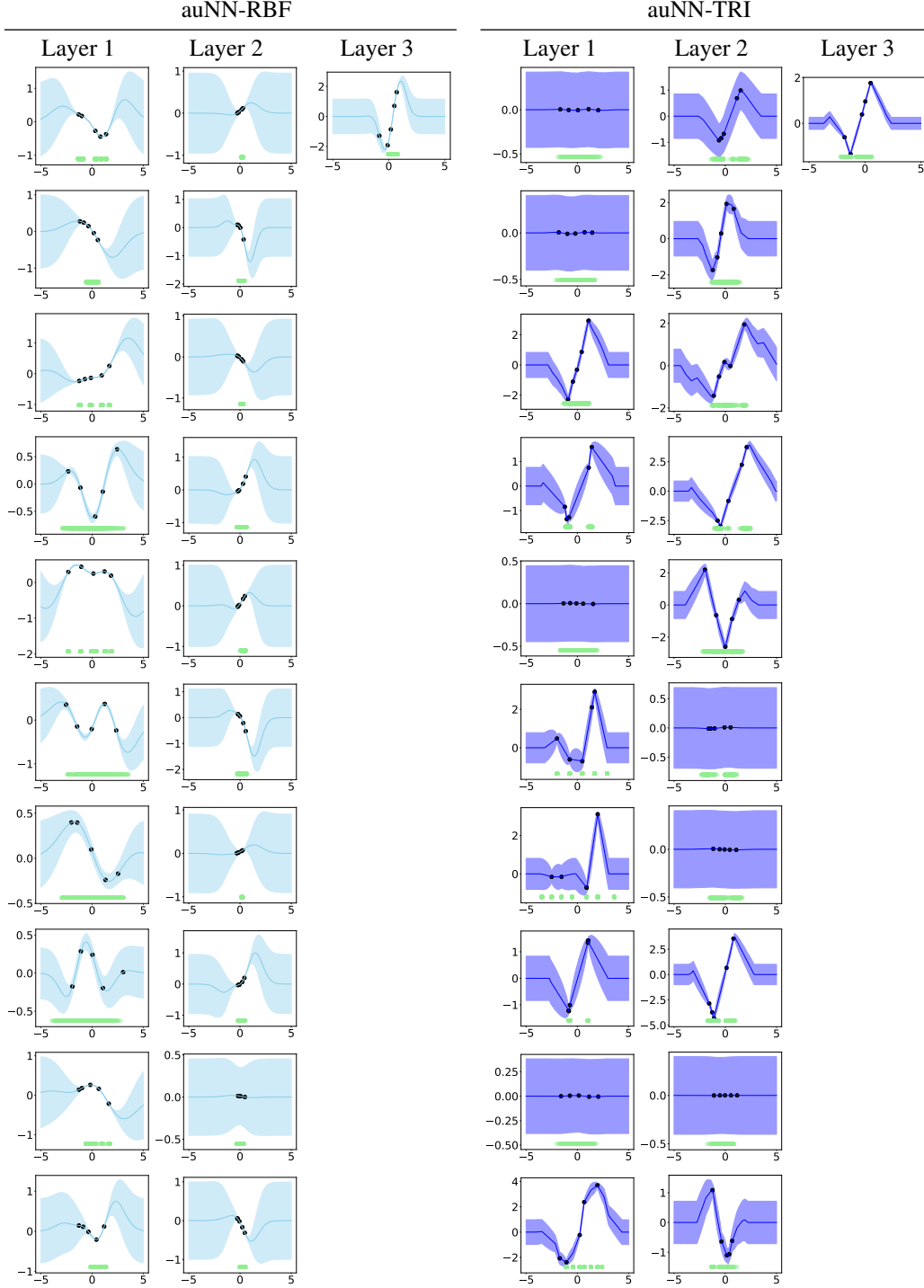


Figure 8: A complete example of the activation functions learned by auNN with RBF and TRI kernels. These were obtained for the Energy dataset with the first gap split, using three layers, 10 hidden units per (hidden) layer, and 5 inducing points in each unit. Whereas auNN-RBF learns smoother activations, auNN-TRI ones are piece-wise linear, inspired by the ReLu. Notice that auNN allows units to switch off if they are not required. Black dots represent the five inducing points in each unit. Green points are the locations of the input data when propagated to the corresponding unit.

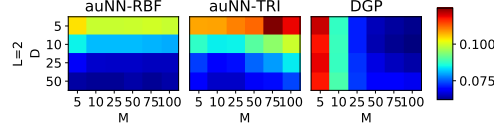


Figure 9: Test RMSE of auNN and DGP for different values of M (number of inducing points) and D (number of hidden units). Results are the average over 5 independent runs on the UCI Kin8 dataset. The lower the better. Whereas DGP improves “by columns” (i.e. with M), auNN does it “by rows” (i.e. with D). This is as theoretically expected, and it is convenient from a scalability viewpoint.

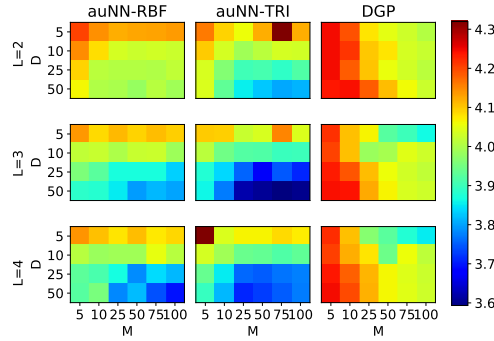


Figure 10: Test RMSE with increasing depth ($L = 2, 3, 4$). This supports that auNN might benefit more than DGP from deeper networks. Moreover, the aforementioned different influence of M and D on DGP and auNN is confirmed here.

Table 9: Test NLL of auNN and DGP for different values of M (number of inducing points) and D (number of hidden units). Mean and one standard error over 5 independent runs on the UCI Kin8 dataset are shown. The lower the better.

D	M	auNN-RBF						auNN-TRI						DGP					
		5	10	25	50	75	100	5	10	25	50	75	100	5	10	25	50	75	100
5		-0.85±0.01	-0.89±0.01	-0.89±0.01	-0.89±0.01	-0.89±0.01	-0.90±0.01	-0.78±0.00	-0.79±0.03	-0.78±0.05	-0.77±0.04	-0.67±0.06	-0.71±0.04	-0.67±0.01	-0.98±0.00	-1.19±0.01	-1.30±0.01	-1.33±0.01	-1.34±0.01
10		-1.06±0.01	-1.09±0.01	-1.09±0.01	-1.09±0.01	-1.10±0.02	-1.10±0.01	-0.96±0.01	-1.02±0.01	-1.03±0.01	-0.98±0.03	-0.94±0.03	-0.89±0.03	-0.69±0.01	-0.98±0.00	-1.19±0.00	-1.30±0.01	-1.33±0.01	-1.35±0.01
25		-1.27±0.02	-1.30±0.02	-1.30±0.02	-1.30±0.02	-1.31±0.01	-1.31±0.02	-1.09±0.01	-1.19±0.01	-1.22±0.01	-1.15±0.02	-1.11±0.01	-1.06±0.03	-0.68±0.01	-0.98±0.00	-1.17±0.01	-1.26±0.01	-1.29±0.01	-1.30±0.01
50		-1.33±0.01	-1.34±0.01	-1.34±0.02	-1.33±0.01	-1.34±0.02	-1.32±0.03	-1.15±0.01	-1.24±0.01	-1.29±0.01	-1.26±0.01	-1.24±0.02	-1.19±0.02	-0.69±0.01	-0.96±0.01	-1.16±0.01	-1.21±0.01	-1.22±0.01	-1.24±0.01

Table 10: Test RMSE of auNN and DGP for different values of M (number of inducing points) and D (number of hidden units). Mean and one standard error over 5 independent runs on the UCI Kin8 dataset are shown. The lower the better.

D	M	auNN-RBF						auNN-TRI						DGP					
		5	10	25	50	75	100	5	10	25	50	75	100	5	10	25	50	75	100
5		0.10±0.00	0.10±0.00	0.10±0.00	0.10±0.00	0.10±0.00	0.10±0.00	0.11±0.00	0.11±0.00	0.11±0.01	0.11±0.00	0.12±0.01	0.12±0.00	0.12±0.00	0.09±0.00	0.07±0.00	0.07±0.00	0.06±0.00	0.06±0.00
10		0.08±0.00	0.08±0.00	0.08±0.00	0.08±0.00	0.08±0.00	0.08±0.00	0.09±0.00	0.08±0.00	0.08±0.00	0.09±0.00	0.09±0.00	0.10±0.00	0.12±0.00	0.09±0.00	0.07±0.00	0.06±0.00	0.06±0.00	0.06±0.00
25		0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.08±0.00	0.08±0.00	0.08±0.00	0.12±0.00	0.09±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.06±0.00
50		0.06±0.00	0.06±0.00	0.06±0.00	0.06±0.00	0.06±0.00	0.06±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.12±0.00	0.09±0.00	0.07±0.00	0.07±0.00	0.07±0.00	0.07±0.00

Table 11: Test NLL of auNN and DGP for different values of M (number of inducing points) and D (number of hidden units) as the depth increases from $L = 2$ to $L = 4$. Mean and one standard error over 5 independent runs on the UCI Power dataset are shown. The lower the better.

L	D	M	auNN-RBF						auNN-TRI						DGP					
			5	10	25	50	75	100	5	10	25	50	75	100	5	10	25	50	75	100
2	5		2.86±0.02	2.84±0.02	2.84±0.02	2.84±0.02	2.84±0.02	2.84±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.84±0.02	2.89±0.03	2.84±0.02	2.87±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.81±0.02	2.81±0.02
	10		2.84±0.02	2.83±0.02	2.82±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.84±0.02	2.82±0.02	2.80±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.87±0.02	2.86±0.02	2.83±0.02	2.83±0.02	2.82±0.02	2.81±0.02
	25		2.83±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.83±0.02	2.80±0.02	2.78±0.02	2.78±0.02	2.78±0.02	2.79±0.02	2.87±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.81±0.02	2.81±0.02
	50		2.82±0.02	2.81±0.02	2.81±0.02	2.80±0.02	2.81±0.02	2.81±0.02	2.82±0.02	2.80±0.02	2.77±0.02	2.76±0.02	2.76±0.02	2.76±0.03	2.86±0.02	2.87±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.81±0.02
3	5		2.84±0.02	2.83±0.02	2.83±0.02	2.83±0.03	2.83±0.02	2.83±0.02	2.84±0.02	2.83±0.02	2.82±0.02	2.82±0.02	2.85±0.02	2.82±0.02	2.86±0.02	2.83±0.02	2.82±0.02	2.79±0.02	2.78±0.02	2.77±0.01
	10		2.81±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.81±0.02	2.80±0.02	2.82±0.02	2.80±0.02	2.79±0.02	2.79±0.02	2.78±0.02	2.78±0.02	2.86±0.02	2.83±0.02	2.80±0.02	2.81±0.02	2.82±0.02	2.81±0.02
	25		2.80±0.02	2.79±0.02	2.77±0.02	2.77±0.02	2.77±0.02	2.77±0.02	2.79±0.02	2.77±0.02	2.74±0.02	2.72±0.02	2.74±0.03	2.74±0.03	2.86±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.81±0.02	2.81±0.02
	50		2.78±0.02	2.78±0.02	2.77±0.02	2.76±0.02	2.76±0.02	2.76±0.03	2.78±0.02	2.75±0.02	2.71±0.02	2.71±0.03	2.70±0.03	2.70±0.02	2.87±0.02	2.87±0.02	2.84±0.02	2.82±0.02	2.82±0.02	2.81±0.02
4	5		2.84±0.02	2.83±0.02	2.82±0.02	2.83±0.02	2.82±0.01	2.83±0.02	3.69±0.35	2.83±0.01	2.83±0.02	2.83±0.02	2.83±0.02	2.82±0.02	2.86±0.02	2.83±0.02	2.80±0.02	2.79±0.02	2.78±0.02	2.77±0.02
	10		2.81±0.02	2.80±0.02	2.80±0.02	2.80±0.02	2.82±0.02	2.81±0.02	2.83±0.02	2.81±0.01	2.79±0.01	2.79±0.02	2.79±0.02	2.79±0.02	2.86±0.02	2.84±0.02	2.83±0.02	2.79±0.02	2.82±0.02	2.81±0.02
	25		2.79±0.02	2.78±0.02	2.77±0.02	2.75±0.02	2.77±0.02	2.76±0.02	2.80±0.01	2.78±0.02	2.75±0.02	2.74±0.02	2.75±0.03	2.75±0.02	2.86±0.02	2.85±0.02	2.83±0.02	2.82±0.02	2.82±0.02	2.81±0.02
	50		2.79±0.02	2.80±0.02	2.75±0.03	2.77±0.03	2.75±0.03	2.74±0.03	2.79±0.01	2.77±0.01	2.73±0.02	2.74±0.02	2.74±0.02	2.75±0.02	2.87±0.02	2.85±0.02	2.84±0.02	2.82±0.02	2.82±0.02	2.81±0.02